

helps reduce consultation time with the new doctor," says Santra.

The doctor usually reverts to the RMP with any of four messages: 'Nothing to worry,' 'Change the drug,' 'Change dosage' or 'Bring patient to the hospital'.

Another feature that makes iKure kiosks stand out is their ability to alert doctors in the case of an emergency. This is done through an application programming interface (API) called 'Rules Engine,' which makes the kiosks intelligent enough to figure out when a patient requires immediate attention and alert a doctor.

The Rules Engine sits in the server and analyses every piece of new data that is fed into the database. It processes the data and compares it with the data already stored. If the data is within the normal limits, it considers the case to be worthy of escalation. If the patient's reading crosses this limit, an alert SMS is sent to a registered number assigned to the pool of doctors who are part of this service. The available doctor can then log on to the server using his mobile phone or computer and connect with the practitioner stationed at the kiosk.

Key technological hurdles

Developing such a system was not without challenges. Santra enumerates the key ones: "Many medical instruments available in the market output data in the analogue format. There were many hiccups at the stage of configuring instruments with WHIMS. Every medical equipment used for measuring data—digital or



Model kiosk for city

analogue—behaves differently. We took up the challenge and cracked how the data can be captured, processed and transmitted. Every instrument had to be configured to enable automatic digital transmission of data through Zigbee, Wi-Fi or any other available network."

Another challenge related to transmission of ECG reports as data volume is huge for these reports. Santra tells how this was overcome: "To transmit such reports we optimise the data and send minimal information. A technique based on AZTEC technology is used for optimisation and compression of ECG signals. This enables the data to be sent through a 330-character-long SMS. iKure owns a patent for this technology."

iKure also holds patents for the position detection technique used to tap data from a mercury sphygmomanometer (*aka* blood pressure meter) by

using the metallic property of mercury, WHIMS and Rules Engine.

Going forward...

In the next six months, iKure, in collaboration with global partners such as Global Venture Lab, Finland, plans to increase the number of its health kiosks to 100, and probably to 500 in the next two years, reveals Santra. The organisation also plans to launch health kiosks for homes, which will especially help senior citizens who are ill and find it difficult to visit a doctor.

The technology used in the kiosk will be similar to rural health kiosks but it would not include a medical practitioner. Measurements will be sent directly to the server, from which a doctor will be able to access this data and diagnose.

"On the technology front, we would like to use devices that are more sophisticated and transmit a huge volume of data as is generated through CT scan or MRI," says Santra. iKure also plans to expand to rural areas of other states and even cities through company-owned, franchise-model health kiosks.

Over time, the initiative may help in taking healthcare to people residing in the remotest nooks and corners of India, where access to even the basic healthcare facilities is still a far-fetched dream! ●

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The need that iKure addresses

A research conducted by iKure shows that a patient can travel only up to 65 km in the event of an emergency, and in most cases, the nearest hospital is more than 300 km away. For every 1600 patients, there is just one doctor.

"The main issues in rural regions are infant mortality, maternal death, life expectancy at birth and prenatal mortality. It is shocking to learn that over 75 per cent of such cases could have been avoided with basic healthcare and amenities. iKure health kiosks are a step towards addressing this need," says Sujay Santra, founder-director, iKure Techsoft. iKure does research for remote monitoring, teleradiology, hosted hospital information system, telepathology and mobile technology in health care. It works closely with IIT Kharagpur and Aegle Angels Foundation for these technology research projects.